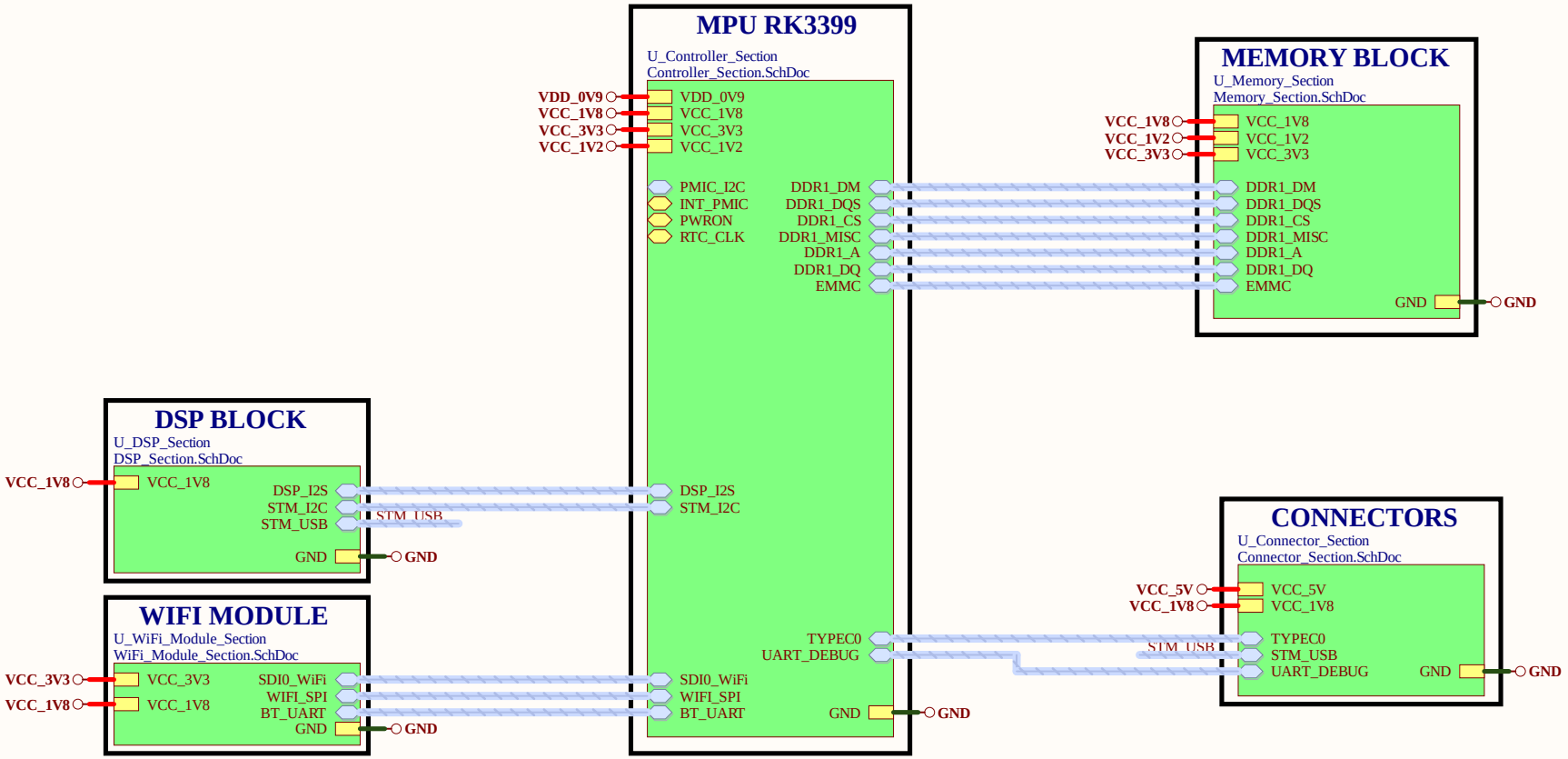
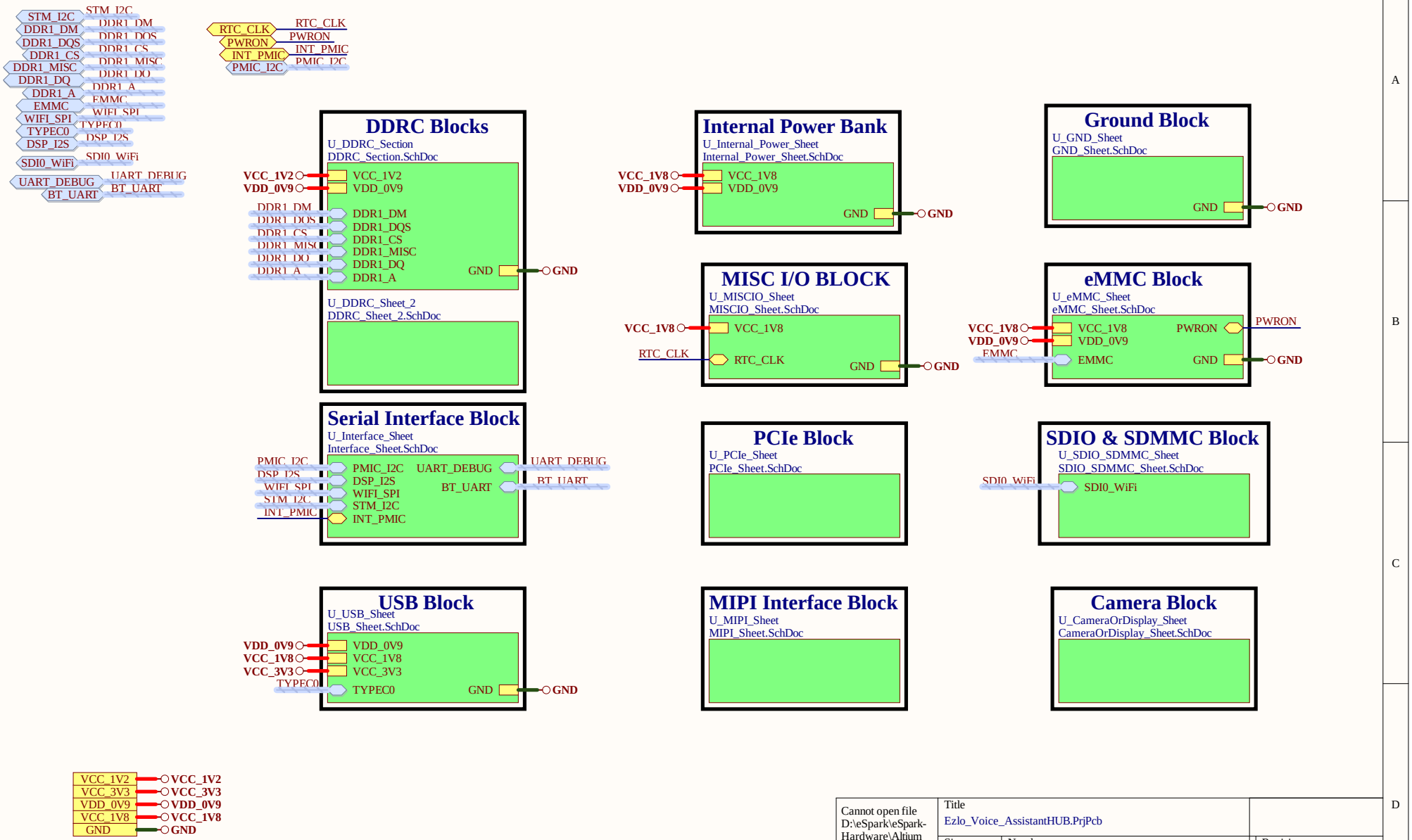
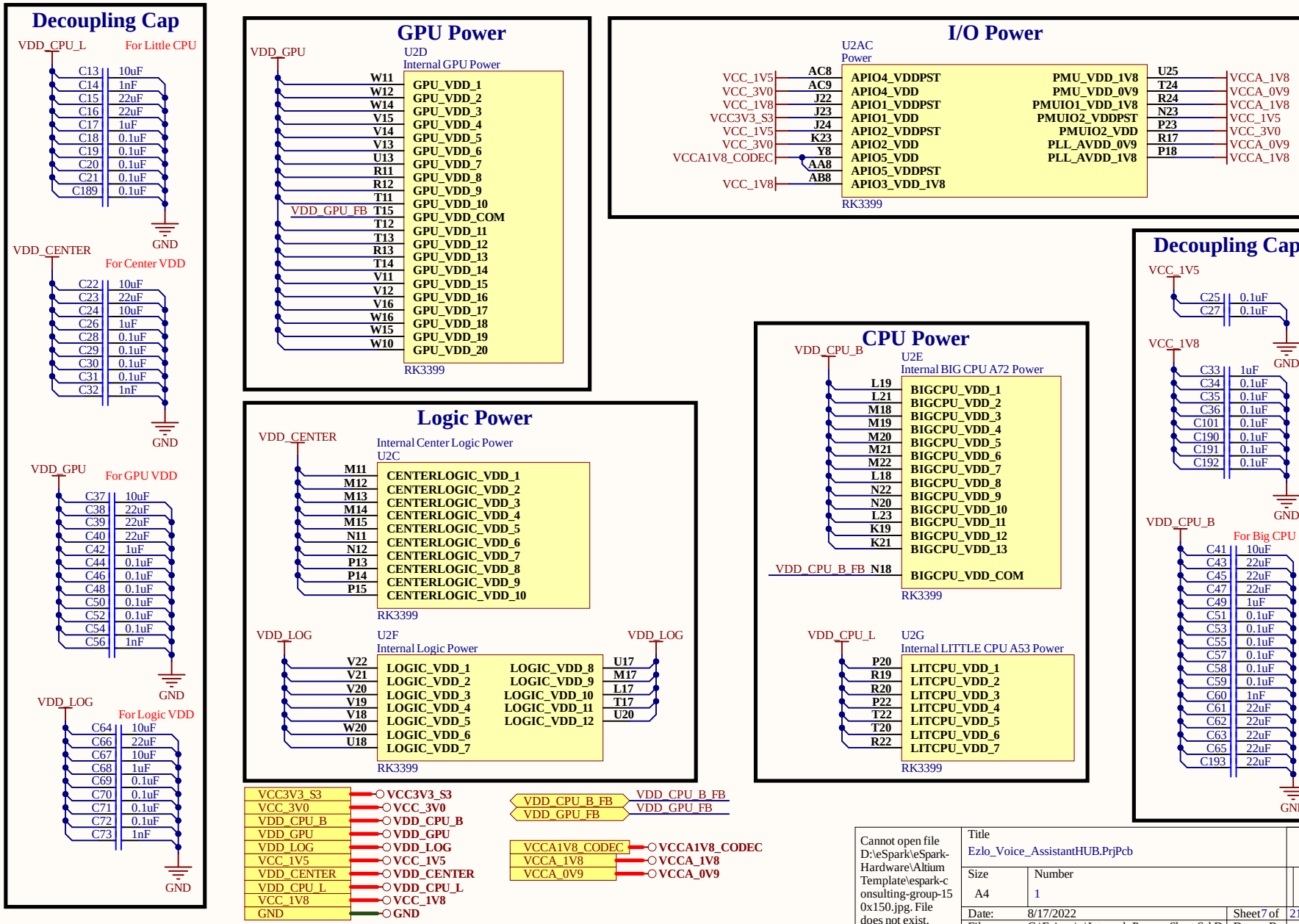
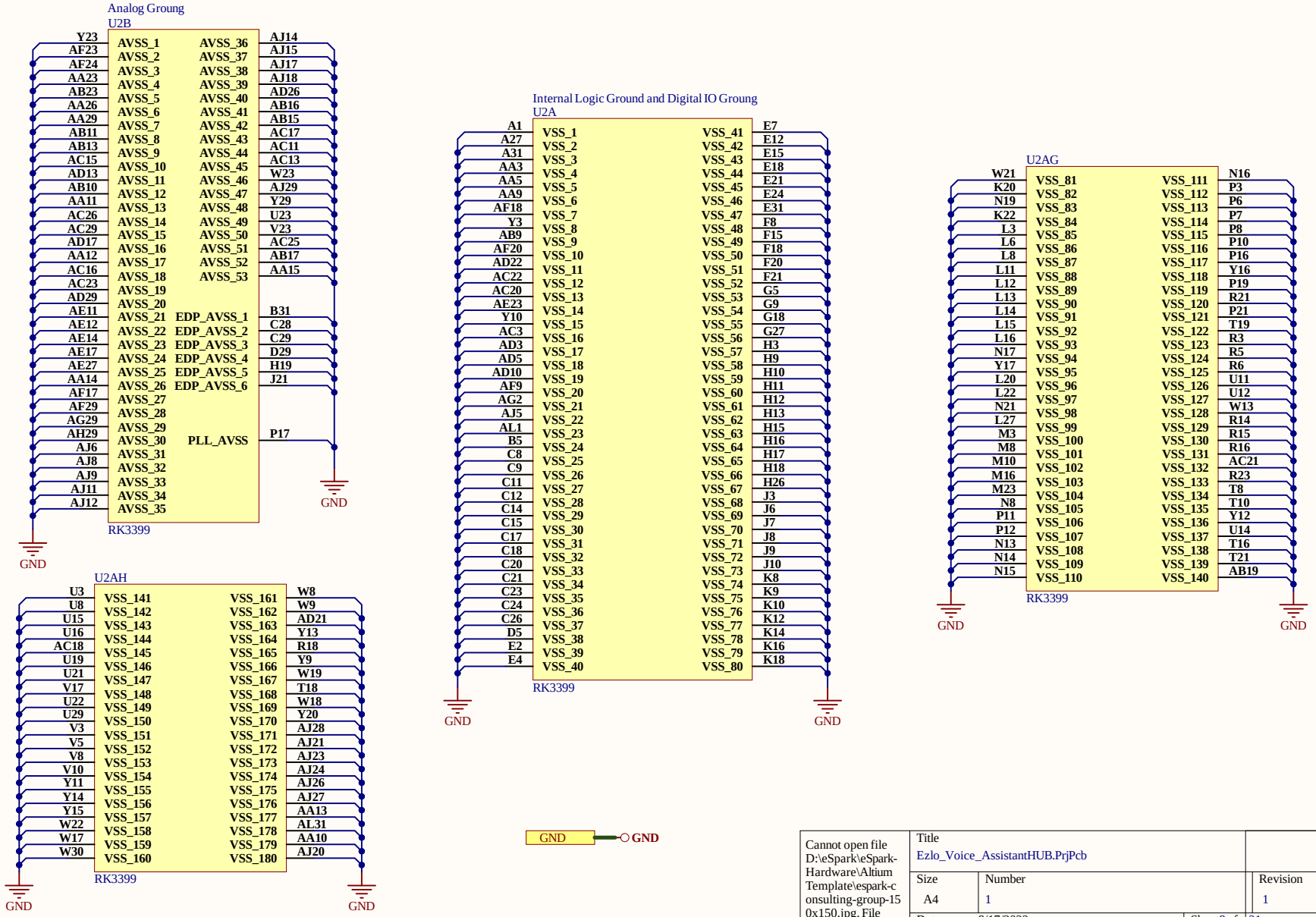


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	Size: A4	Number: 1	Revision: 1
	Date: 8/17/2022	Sheet 1 of 21	
	File: C:\Faizan\..\Top_Sheet.SchDoc	Drawn By: Huzaifa Ali	



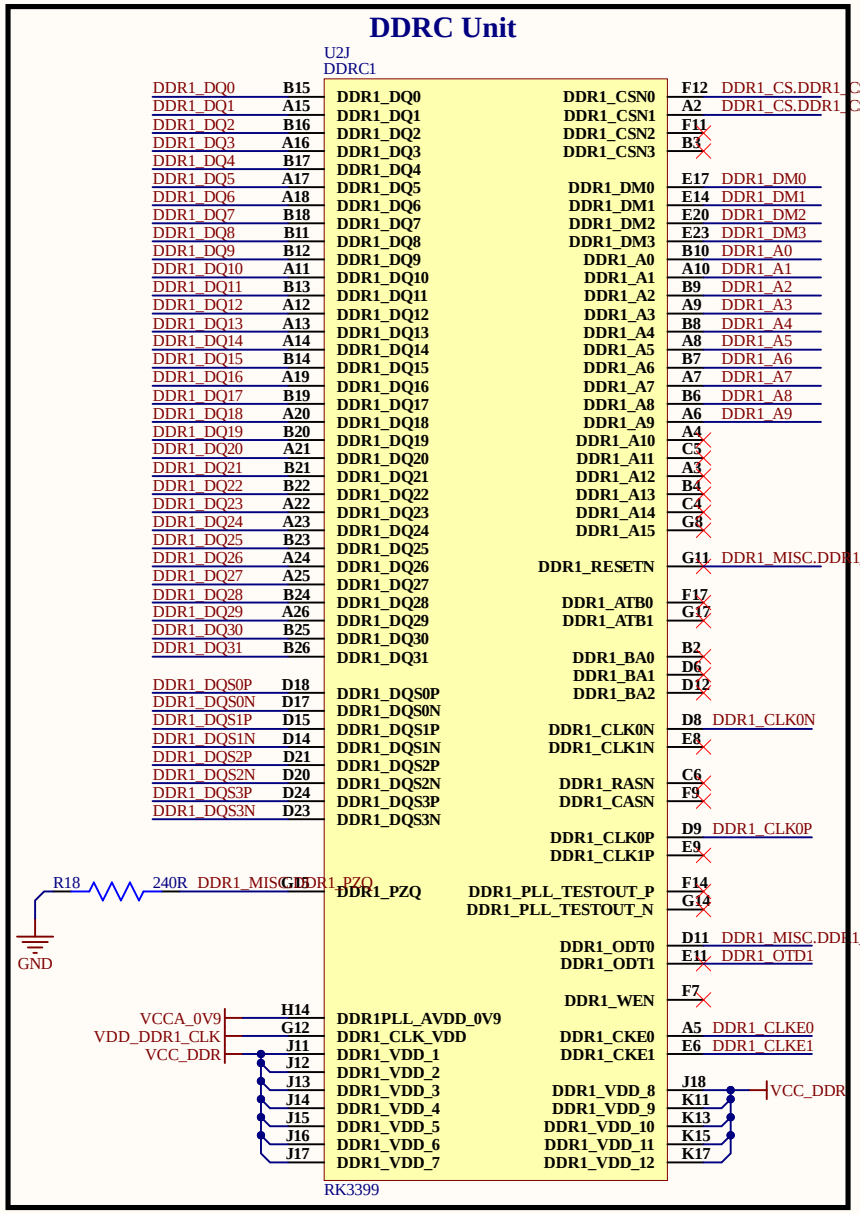
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	Size: A4	Number: 1	Revision: 1
	Date: 8/17/2022	Sheet 6 of 21	
	File: C:\Faizan\..\Controller_Section.SchDoc	Drawn By: Huzaifa Ali	


Decoupling Cap
VDD_CPU_L
For Little CPU
C13 10uF
C14 1nF
C15 22uF
C16 22uF
C17 1uF
C18 0.1uF
C19 0.1uF
C20 0.1uF
C21 0.1uF
C189 0.1uF
GND
VDD_CENTER
For Center VDD
C22 10uF
C23 22uF
C24 10uF
C26 1uF
C28 0.1uF
C29 0.1uF
C30 0.1uF
C31 0.1uF
C32 1nF
GND
VDD_GPU
For GPU VDD
C37 10uF
C38 22uF
C39 22uF
C40 22uF
C42 1uF
C44 0.1uF
C46 0.1uF
C48 0.1uF
C50 0.1uF
C52 0.1uF
C54 0.1uF
C56 1nF
GND
VDD_LOG
For Logic VDD
C64 10uF
C66 22uF
C67 10uF
C68 1uF
C69 0.1uF
C70 0.1uF
C71 0.1uF
C72 0.1uF
C73 1nF
GND
GPU Power
VDD_GPU
U2D
Internal GPU Power
W11 GPU_VDD_1
W12 GPU_VDD_2
W14 GPU_VDD_3
V15 GPU_VDD_4
V14 GPU_VDD_5
V13 GPU_VDD_6
U13 GPU_VDD_7
R11 GPU_VDD_8
R12 GPU_VDD_9
T11 GPU_VDD_10
VDD_GPU_FB T15 GPU_VDD_COM
T12 GPU_VDD_11
T13 GPU_VDD_12
R13 GPU_VDD_13
T14 GPU_VDD_14
V11 GPU_VDD_15
V12 GPU_VDD_16
V16 GPU_VDD_17
W16 GPU_VDD_18
W15 GPU_VDD_19
W10 GPU_VDD_20
RK3399
I/O Power
U2AC
Power
VCC_1V5 AC8
VCC_3V0 AC9
VCC_1V8 J22
VCC3V3_S3 J23
VCC_1V5 J24
VCC_3V0 K23
VCCA1V8_CODEC Y8
AA8
VCC_1V8 AB8
APIO4_VDDPST
APIO4_VDD
APIO1_VDDPST
APIO1_VDD
APIO2_VDDPST
APIO2_VDD
APIO5_VDD
APIO5_VDDPST
APIO3_VDD_1V8
PMU_VDD_1V8 U25
PMU_VDD_0V9 T24
PMUIO1_VDD_1V8 R24
PMUIO2_VDDPST N23
PMUIO2_VDD P23
PLL_AVDD_0V9 R17
PLL_AVDD_1V8 P18
VCCA_1V8
VCCA_0V9
VCCA_1V8
VCC_1V5
VCC_3V0
VCCA_0V9
VCCA_1V8
RK3399
Logic Power
VDD_CENTER
Internal Center Logic Power
U2C
M11 CENTERLOGIC_VDD_1
M12 CENTERLOGIC_VDD_2
M13 CENTERLOGIC_VDD_3
M14 CENTERLOGIC_VDD_4
M15 CENTERLOGIC_VDD_5
N11 CENTERLOGIC_VDD_6
N12 CENTERLOGIC_VDD_7
P13 CENTERLOGIC_VDD_8
P14 CENTERLOGIC_VDD_9
P15 CENTERLOGIC_VDD_10
RK3399
VDD_LOG
U2F
Internal Logic Power
V22 LOGIC_VDD_1
V21 LOGIC_VDD_2
V20 LOGIC_VDD_3
V19 LOGIC_VDD_4
V18 LOGIC_VDD_5
W20 LOGIC_VDD_6
U18 LOGIC_VDD_7
LOGIC_VDD_8 U17
LOGIC_VDD_9 M17
LOGIC_VDD_10 L17
LOGIC_VDD_11 T17
LOGIC_VDD_12 U20
VDD_LOG
RK3399
CPU Power
VDD_CPU_B
U2E
Internal BIG CPU A72 Power
L19 BIGCPU_VDD_1
L21 BIGCPU_VDD_2
M18 BIGCPU_VDD_3
M19 BIGCPU_VDD_4
M20 BIGCPU_VDD_5
M21 BIGCPU_VDD_6
M22 BIGCPU_VDD_7
L18 BIGCPU_VDD_8
N22 BIGCPU_VDD_9
N20 BIGCPU_VDD_10
L23 BIGCPU_VDD_11
K19 BIGCPU_VDD_12
K21 BIGCPU_VDD_13
VDD_CPU_B_FB N18 BIGCPU_VDD_COM
RK3399
VDD_CPU_L
U2G
Internal LITTLE CPU A53 Power
P20 LITCPU_VDD_1
R19 LITCPU_VDD_2
R20 LITCPU_VDD_3
P22 LITCPU_VDD_4
T22 LITCPU_VDD_5
T20 LITCPU_VDD_6
R22 LITCPU_VDD_7
RK3399
Decoupling Cap
VCC_1V5
C25 0.1uF
C27 0.1uF
GND
VCC_1V8
C33 1uF
C34 0.1uF
C35 0.1uF
C36 0.1uF
C101 0.1uF
C190 0.1uF
C191 0.1uF
C192 0.1uF
GND
VDD_CPU_B
For Big CPU
C41 10uF
C43 22uF
C45 22uF
C47 22uF
C49 1uF
C51 0.1uF
C53 0.1uF
C55 0.1uF
C57 0.1uF
C58 0.1uF
C59 0.1uF
C60 1nF
C61 22uF
C62 22uF
C63 22uF
C65 22uF
C193 22uF
GND
VCC3V3_S3
VCC_3V0
VDD_CPU_B
VDD_GPU
VDD_LOG
VCC_1V5
VDD_CENTER
VDD_CPU_L
VCC_1V8
GND
VDD_CPU_B_FB
VDD_GPU_FB
VCCA1V8_CODEC
VCCA_1V8
VCCA_0V9
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
Revision 1
Date: 8/17/2022
Sheet 7 of 21
File: C:\Faizan\..\Internal_Power_Sheet.SchDoc
Drawn By: Huzaifa Ali



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Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	8/17/2022	Sheet 8 of	21
File:	C:\Faizan\..\GND_Sheet.SchDoc	Drawn By:	Huzaifa Ali


DDRC Unit
U2J
DDRC1
DDR1_DQ0 B15
DDR1_DQ1 A15
DDR1_DQ2 B16
DDR1_DQ3 A16
DDR1_DQ4 B17
DDR1_DQ5 A17
DDR1_DQ6 A18
DDR1_DQ7 B18
DDR1_DQ8 B11
DDR1_DQ9 B12
DDR1_DQ10 A11
DDR1_DQ11 B13
DDR1_DQ12 A12
DDR1_DQ13 A13
DDR1_DQ14 A14
DDR1_DQ15 B14
DDR1_DQ16 A19
DDR1_DQ17 B19
DDR1_DQ18 A20
DDR1_DQ19 B20
DDR1_DQ20 A21
DDR1_DQ21 B21
DDR1_DQ22 B22
DDR1_DQ23 A22
DDR1_DQ24 A23
DDR1_DQ25 B23
DDR1_DQ26 A24
DDR1_DQ27 A25
DDR1_DQ28 B24
DDR1_DQ29 A26
DDR1_DQ30 B25
DDR1_DQ31 B26
DDR1_DQS0P D18
DDR1_DQS0N D17
DDR1_DQS1P D15
DDR1_DQS1N D14
DDR1_DQS2P D21
DDR1_DQS2N D20
DDR1_DQS3P D24
DDR1_DQS3N D23
R18 240R DDR1_MISC.DDR1_PZQ
GND
VCCA_0V9 H14
VDD_DDR1_CLK G12
VCC_DDR J11
J12
J13
J14
J15
J16
J17
DDR1_DQ0
DDR1_DQ1
DDR1_DQ2
DDR1_DQ3
DDR1_DQ4
DDR1_DQ5
DDR1_DQ6
DDR1_DQ7
DDR1_DQ8
DDR1_DQ9
DDR1_DQ10
DDR1_DQ11
DDR1_DQ12
DDR1_DQ13
DDR1_DQ14
DDR1_DQ15
DDR1_DQ16
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DDR1_DQ25
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DDR1_DQ27
DDR1_DQ28
DDR1_DQ29
DDR1_DQ30
DDR1_DQ31
DDR1_DQS0P
DDR1_DQS0N
DDR1_DQS1P
DDR1_DQS1N
DDR1_DQS2P
DDR1_DQS2N
DDR1_DQS3P
DDR1_DQS3N
DDR1PLL_AVDD_0V9
DDR1_CLK_VDD
DDR1_VDD_1
DDR1_VDD_2
DDR1_VDD_3
DDR1_VDD_4
DDR1_VDD_5
DDR1_VDD_6
DDR1_VDD_7
DDR1_CSN0 F12 DDR1_CS.DDR1_CSN0
DDR1_CSN1 A2 DDR1_CS.DDR1_CSN1
DDR1_CSN2 F11
DDR1_CSN3 B3
DDR1_DM0 E17 DDR1_DM0
DDR1_DM1 E14 DDR1_DM1
DDR1_DM2 E20 DDR1_DM2
DDR1_DM3 E23 DDR1_DM3
DDR1_A0 B10 DDR1_A0
DDR1_A1 A10 DDR1_A1
DDR1_A2 B9 DDR1_A2
DDR1_A3 A9 DDR1_A3
DDR1_A4 B8 DDR1_A4
DDR1_A5 A8 DDR1_A5
DDR1_A6 B7 DDR1_A6
DDR1_A7 A7 DDR1_A7
DDR1_A8 B6 DDR1_A8
DDR1_A9 A6 DDR1_A9
DDR1_A10 A4
DDR1_A11 C5
DDR1_A12 A3
DDR1_A13 B4
DDR1_A14 C4
DDR1_A15 G8
DDR1_RESETN G11 DDR1_MISC.DDR1_RST
DDR1_ATB0 F17
DDR1_ATB1 G17
DDR1_BA0 B2
DDR1_BA1 D6
DDR1_BA2 D12
DDR1_CLK0N D8 DDR1_CLK0N
DDR1_CLK1N E8
DDR1_RASN C6
DDR1_CASN F9
DDR1_CLK0P D9 DDR1_CLK0P
DDR1_CLK1P E9
DDR1_PZQ G15
DDR1_PLL_TESTOUT_P F14
DDR1_PLL_TESTOUT_N G14
DDR1_ODT0 D11 DDR1_MISC.DDR1_OTD
DDR1_ODT1 E11 DDR1_OTD1
DDR1_WEN F7
DDR1_CKE0 A5 DDR1_CKE0
DDR1_CKE1 E6 DDR1_CKE1
DDR1_VDD_8 J18
DDR1_VDD_9 K11
DDR1_VDD_10 K13
DDR1_VDD_11 K15
DDR1_VDD_12 K17
VCC_DDR
RK3399
DDR1_A
DDR1_A0 DDR1_A0
DDR1_A1 DDR1_A1
DDR1_A2 DDR1_A2
DDR1_A3 DDR1_A3
DDR1_A4 DDR1_A4
DDR1_A5 DDR1_A5
DDR1_A6 DDR1_A6
DDR1_A7 DDR1_A7
DDR1_A8 DDR1_A8
DDR1_A9 DDR1_A9
DDR1_DQ
DDR1_DQ0 DDR1_DQ0
DDR1_DQ1 DDR1_DQ1
DDR1_DQ2 DDR1_DQ2
DDR1_DQ3 DDR1_DQ3
DDR1_DQ4 DDR1_DQ4
DDR1_DQ5 DDR1_DQ5
DDR1_DQ6 DDR1_DQ6
DDR1_DQ7 DDR1_DQ7
DDR1_DQ8 DDR1_DQ8
DDR1_DQ9 DDR1_DQ9
DDR1_DQ10 DDR1_DQ10
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DDR1_DQ23 DDR1_DQ23
DDR1_DQ24 DDR1_DQ24
DDR1_DQ25 DDR1_DQ25
DDR1_DQ26 DDR1_DQ26
DDR1_DQ27 DDR1_DQ27
DDR1_DQ28 DDR1_DQ28
DDR1_DQ29 DDR1_DQ29
DDR1_DQ30 DDR1_DQ30
DDR1_DQ31 DDR1_DQ31
Test Points
DDR1_MISC.DDR1_OTD P33
DDR1_OTD1 P34
Decoupling Cap
VCC_DDR VCCA_0V9
C102 0.1uF
C103 22uF
C104 10uF
C8 1uF
C106 0.1uF
C107 0.1uF
C108 0.1uF
C109 0.1uF
C110 0.1uF
C111 0.1uF
C112 0.1uF
GND
VDD_DDR1_CLK
R31 0R VCC_DDR
C105 0.1uF GND
DDR1_DQS
DDR1_DQS0P DDR1_DQS0P
DDR1_DQS0N DDR1_DQS0N
DDR1_DQS1P DDR1_DQS1P
DDR1_DQS1N DDR1_DQS1N
DDR1_DQS2P DDR1_DQS2P
DDR1_DQS2N DDR1_DQS2N
DDR1_DQS3P DDR1_DQS3P
DDR1_DQS3N DDR1_DQS3N
DDR1_DM
DDR1_DM0 DDR1_DM0
DDR1_DM1 DDR1_DM1
DDR1_DM2 DDR1_DM2
DDR1_DM3 DDR1_DM3
DDR1_CS
DDR1_CSN0 DDR1_CS.DDR1_CSN0
DDR1_CSN1 DDR1_CS.DDR1_CSN1
DDR1_MISC
DDR1_CLK0N
DDR1_CLK0P
DDR1_PZQ DDR1_MISC.DDR1_PZQ
DDR1_OTD DDR1_MISC.DDR1_OTD
DDR1_CLKE0
DDR1_CLKE1
DDR1_RST DDR1_MISC.DDR1_RST
VCCA_0V9 VCCA_0V9
VCC_DDR VCC_1V2
GND GND
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Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
Revision 1
Date: 8/17/2022
Sheet 9 of 21
File: C:\Faizan\..\DDRC_Section.SchDoc
Drawn By: Huzaifa Ali

U2I
DDRC0

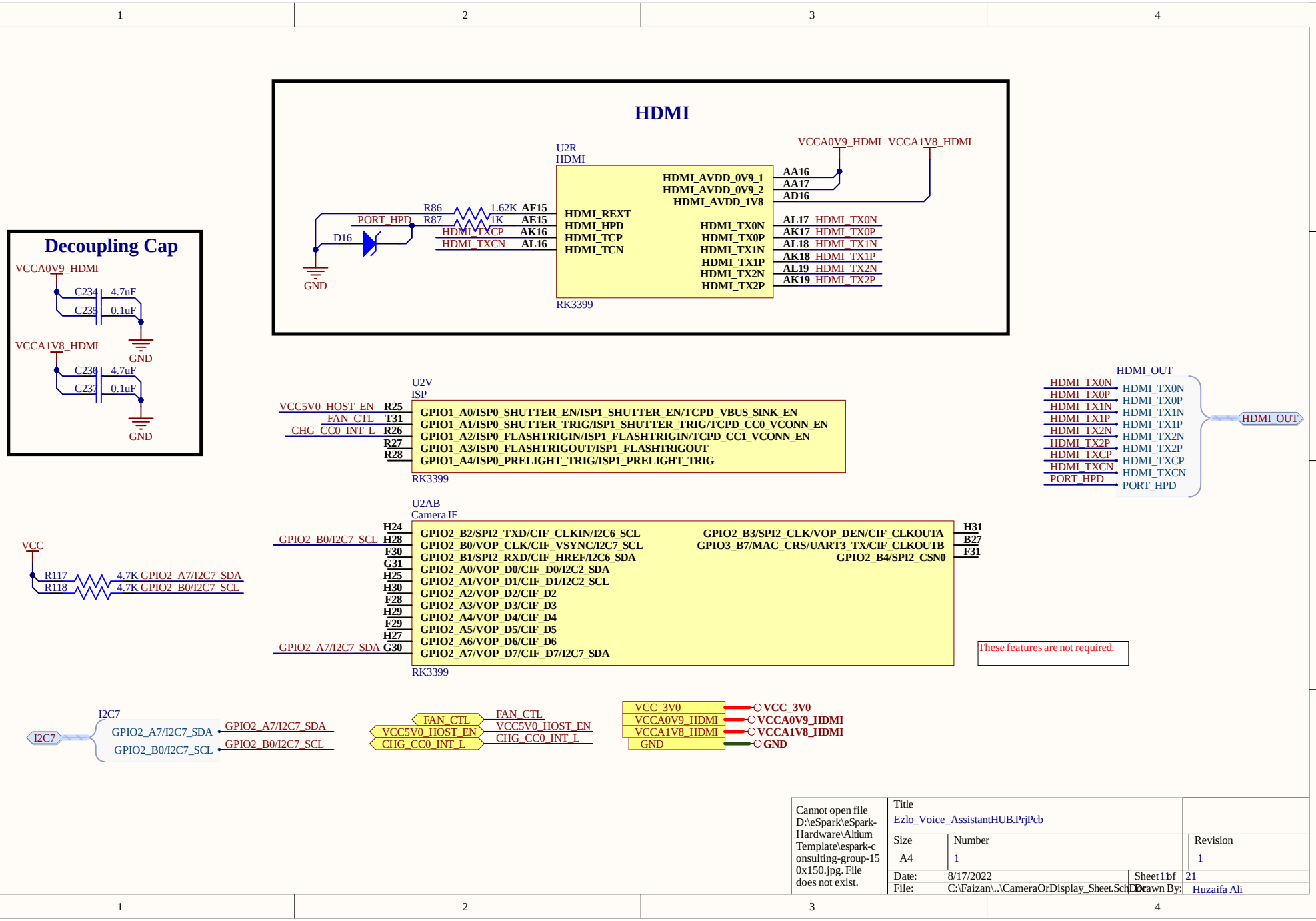
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AB1	DDR0_DQ1	DDR0_CLK1P	H5
AA2	DDR0_DQ2	DDR0_CLK0N	J4
AA1	DDR0_DQ3	DDR0_CLK1N	J5
Y2	DDR0_DQ4		
Y1	DDR0_DQ5	DDR0_RESETN	L7
W1	DDR0_DQ6		
W2	DDR0_DQ7	DDR0_RASN	F3
AF2	DDR0_DQ8	DDR0_CASN	H6
AE2	DDR0_DQ9		
AF1	DDR0_DQ10	DDR0_DM0	Y5
AD2	DDR0_DQ11	DDR0_DM1	AC5
AE1	DDR0_DQ12	DDR0_DM2	U5
AD1	DDR0_DQ13	DDR0_DM3	P5
AC1	DDR0_DQ14		
AC2	DDR0_DQ15	DDR0_CSN0	M6
V1	DDR0_DQ16	DDR0_CSN1	B1
V2	DDR0_DQ17	DDR0_CSN2	M5
U1	DDR0_DQ18	DDR0_CSN3	C2
U2	DDR0_DQ19	DDR0_A0	F2
T1	DDR0_DQ20	DDR0_A1	F1
T2	DDR0_DQ21	DDR0_A2	G1
R2	DDR0_DQ22	DDR0_A3	G2
R1	DDR0_DQ23	DDR0_A4	H2
P1	DDR0_DQ24	DDR0_A5	H1
P2	DDR0_DQ25	DDR0_A6	J1
N1	DDR0_DQ26	DDR0_A7	J2
M1	DDR0_DQ27	DDR0_A8	K2
N2	DDR0_DQ28	DDR0_A9	K1
L1	DDR0_DQ29	DDR0_A10	D1
M2	DDR0_DQ30	DDR0_A11	E3
L2	DDR0_DQ31	DDR0_A12	C1
		DDR0_A13	D2
		DDR0_A14	D3
Y4	DDR0_DQS0P	DDR0_A15	H7
AA4	DDR0_DQS0N		
AC4	DDR0_DQS1P	DDR0_ATB0	U6
AD4	DDR0_DQS1N	DDR0_ATB1	U7
U4	DDR0_DQS2P		
V4	DDR0_DQS2N	DDR0_BA0	C3
P4	DDR0_DQS3P	DDR0_BA1	F4
R4	DDR0_DQS3N	DDR0_BA2	M4
		DDR0_ODT0	L4
		DDR0_ODT1	L5
R7	DDR0_PZQ		
		DDR0_PLL_TESTOUT_P	V6
		DDR0_PLL_TESTOUT_N	V7
		DDR0_WEN	G6
R8	DDR0PLL_AVDD_0V9		
M7	DDR0_CLK_VDD	DDR0_CKE0	E1
L9	DDR0_VDD_1	DDR0_CKE1	F5
L10	DDR0_VDD_2		
M9	DDR0_VDD_3	DDR0_VDD_8	R10
N9	DDR0_VDD_4	DDR0_VDD_9	T9
N10	DDR0_VDD_5	DDR0_VDD_10	U9
P9	DDR0_VDD_6	DDR0_VDD_11	U10
R9	DDR0_VDD_7	DDR0_VDD_12	V9

RK3399

These features are not required.

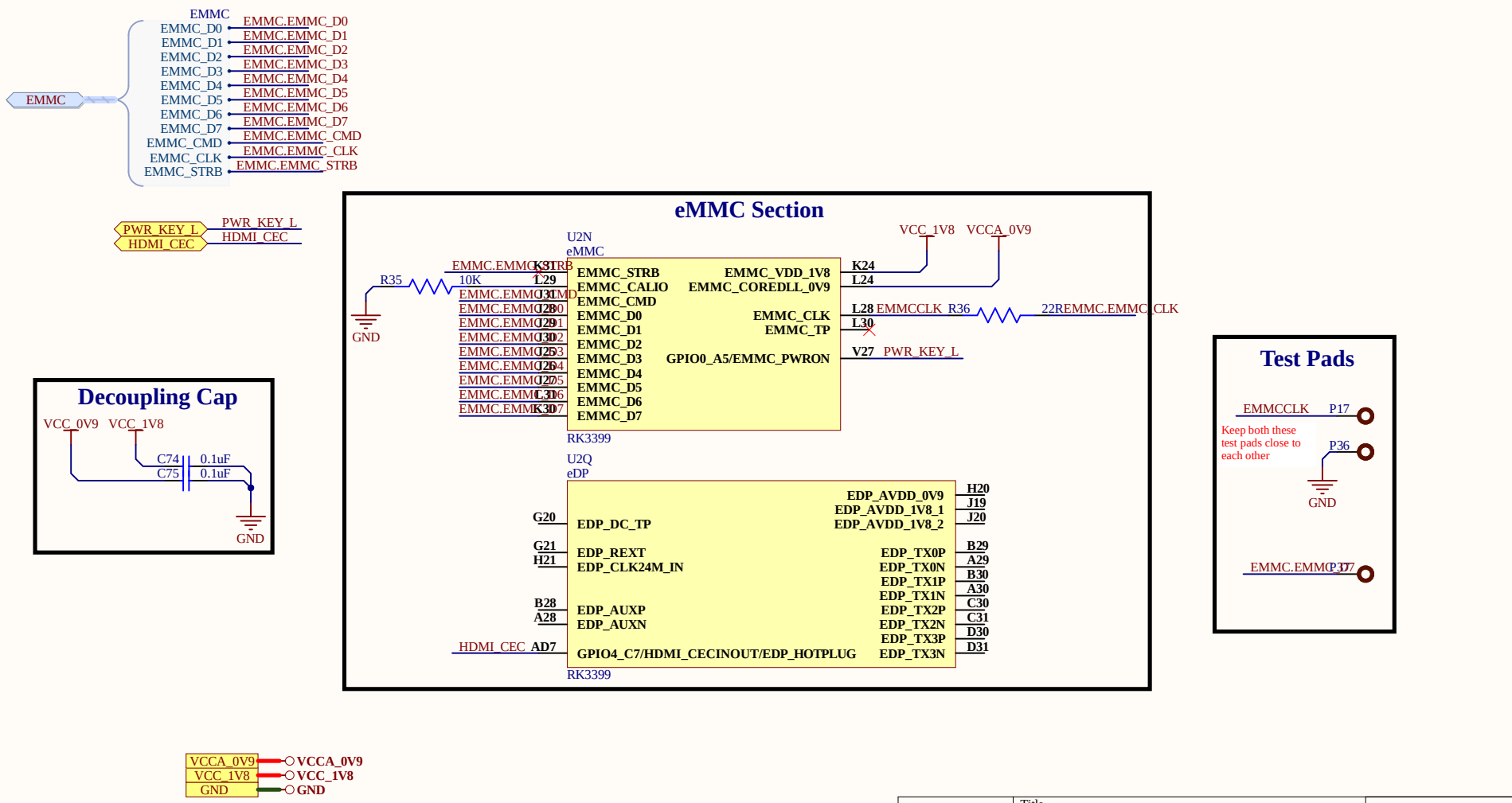
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Size	Number		Revision
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Date:	8/17/2022	Sheet 10 of	21
File:	C:\Faizan\..\DDRC_Sheet_2.SchDoc	Drawn By:	Huzaifa Ali

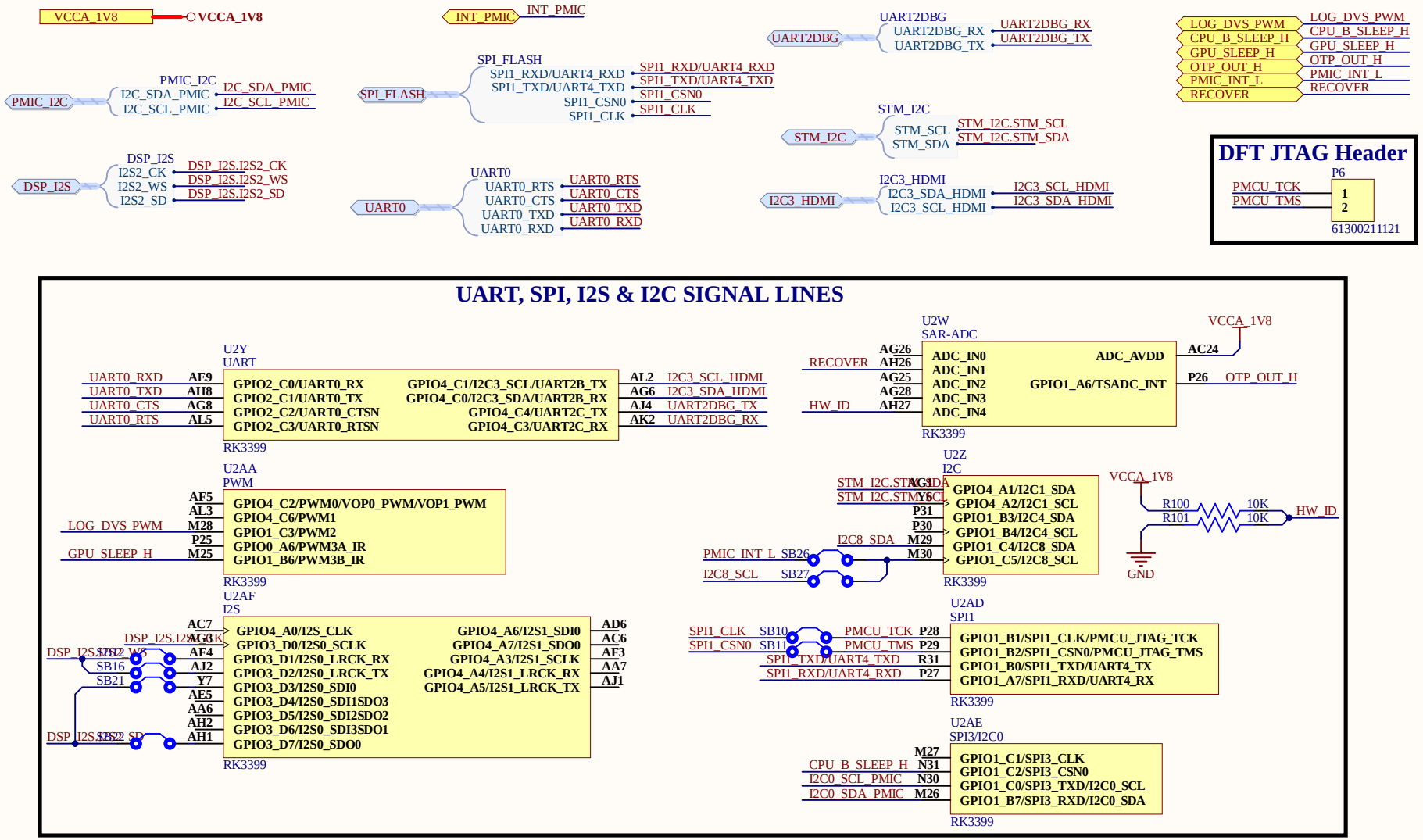


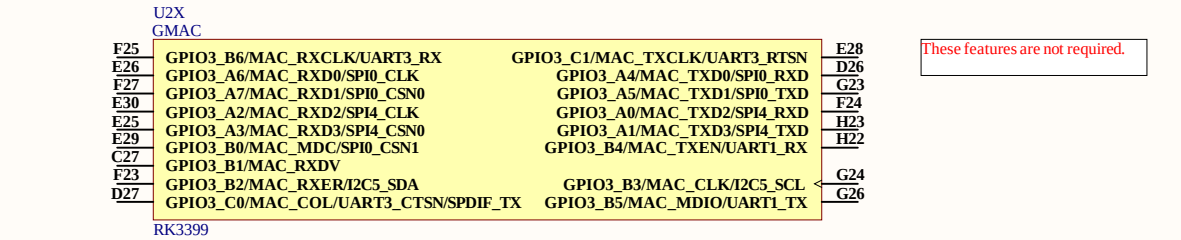
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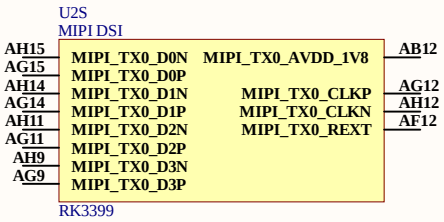
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Size	Number		Revision
A4	1		1
Date:	8/17/2022	Sheet 1 of 21	
File:	C:\Faizan\..\CameraOrDisplay_Sheet.SchDoc	Drawn By:	Huzaifa Ali

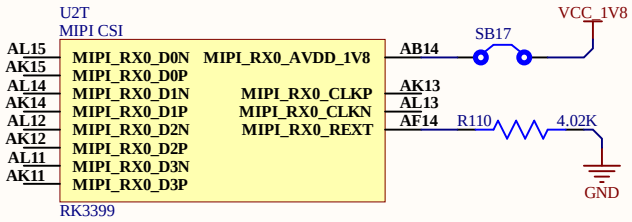


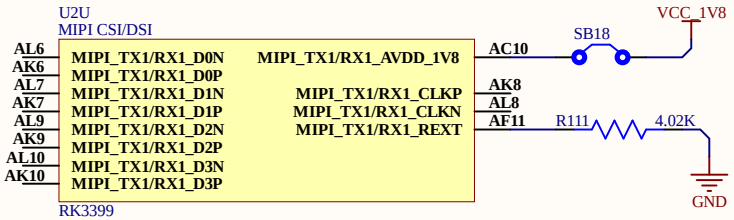
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	Size: A4	Number: 1	Revision: 1
	Date: 8/17/2022	Sheet 12 of 21	
	File: C:\Faizan\..\eMMC_Sheet.SchDoc	Drawn By: Huzaifa Ali	


VCCA_1V8
VCCA_1V8
INT_PMIC
INT_PMIC
UART2DBG
UART2DBG_RX
UART2DBG_TX
UART2DBG_RX
UART2DBG_TX
LOG_DVS_PWM
CPU_B_SLEEP_H
GPU_SLEEP_H
OTP_OUT_H
PMIC_INT_L
RECOVER
PMIC_I2C
I2C_SDA_PMIC
I2C_SCL_PMIC
SPI_FLASH
SPI1_RXD/UART4_RXD
SPI1_TXD/UART4_TXD
SPI1_CSN0
SPI1_CLK
STM_I2C
STM_SCL
STM_SDA
DSP_I2S
I2S2_CK
I2S2_WS
I2S2_SD
UART0
UART0_RTS
UART0_CTS
UART0_TXD
UART0_RXD
I2C3_HDMI
I2C3_SDA_HDMI
I2C3_SCL_HDMI
DFT JTAG Header
P6
PMCU_TCK
PMCU_TMS
61300211121
UART, SPI, I2S & I2C SIGNAL LINES
U2Y
UART
GPIO2_C0/UART0_RX
GPIO2_C1/UART0_TX
GPIO2_C2/UART0_CTSN
GPIO2_C3/UART0_RTSN
GPIO4_C1/I2C3_SCL/UART2B_TX
GPIO4_C0/I2C3_SDA/UART2B_RX
GPIO4_C4/UART2C_TX
GPIO4_C3/UART2C_RX
RK3399
U2W
SAR-ADC
ADC_IN0
ADC_IN1
ADC_IN2
ADC_IN3
ADC_IN4
ADC_AVDD
GPIO1_A6/TSADC_INT
U2AA
PWM
GPIO4_C2/PWM0/VOP0_PWM/VOP1_PWM
GPIO4_C6/PWM1
GPIO1_C3/PWM2
GPIO0_A6/PWM3A_IR
GPIO1_B6/PWM3B_IR
U2Z
I2C
GPIO4_A1/I2C1_SDA
GPIO4_A2/I2C1_SCL
GPIO1_B3/I2C4_SDA
GPIO1_B4/I2C4_SCL
GPIO1_C4/I2C8_SDA
GPIO1_C5/I2C8_SCL
R100
R101
10K
GND
U2AF
I2S
GPIO4_A0/I2S_CLK
GPIO3_D0/I2S0_SCLK
GPIO3_D1/I2S0_LRCK_RX
GPIO3_D2/I2S0_LRCK_TX
GPIO3_D3/I2S0_SDI0
GPIO3_D4/I2S0_SDI1SDO3
GPIO3_D5/I2S0_SDI2SDO2
GPIO3_D6/I2S0_SDI3SDO1
GPIO3_D7/I2S0_SDO0
GPIO4_A6/I2S1_SDI0
GPIO4_A7/I2S1_SDO0
GPIO4_A3/I2S1_SCLK
GPIO4_A4/I2S1_LRCK_RX
GPIO4_A5/I2S1_LRCK_TX
U2AD
SPI1
GPIO1_B1/SPI1_CLK/PMCU_JTAG_TCK
GPIO1_B2/SPI1_CSN0/PMCU_JTAG_TMS
GPIO1_B0/SPI1_TXD/UART4_TX
GPIO1_A7/SPI1_RXD/UART4_RX
U2AE
SPI3/I2C0
GPIO1_C1/SPI3_CLK
GPIO1_C2/SPI3_CSN0
GPIO1_C0/SPI3_TXD/I2C0_SCL
GPIO1_B7/SPI3_RXD/I2C0_SDA
I2C8
I2C8_SCL
I2C8_SDA
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
1
Revision
1
Date:
8/17/2022
Sheet 13 of 21
File:
C:\Faizan\..\Interface_Sheet.SchDoc
Drawn By:
Huzaifa Ali





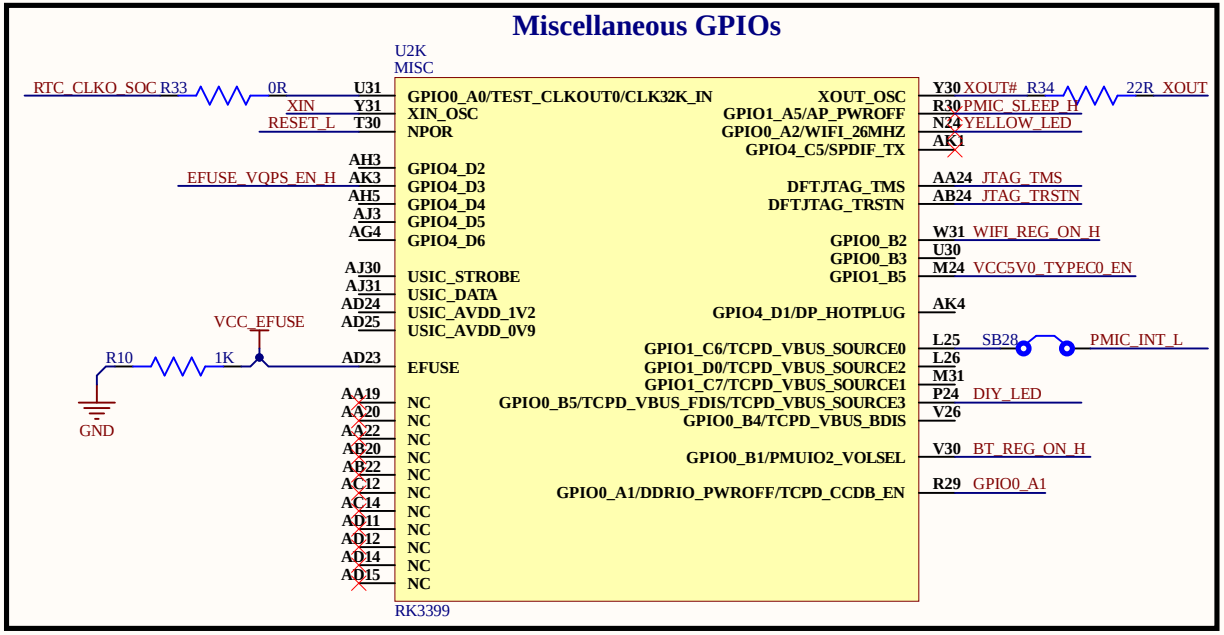




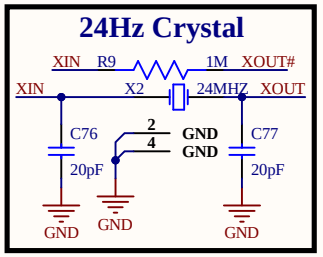


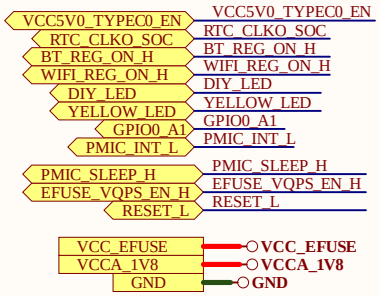
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Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
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Date:	8/17/2022	Sheet 14 of	21
File:	C:\Faizan\..\MIPI_Sheet.SchDoc	Drawn By:	Huzaifa Ali

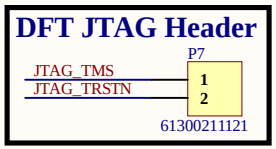


- AP means CPU core in RK3399 including Cortex A72 and Cortex A53.
- MCU and PMCU means built-in micro-controller in RK3399 core domain.





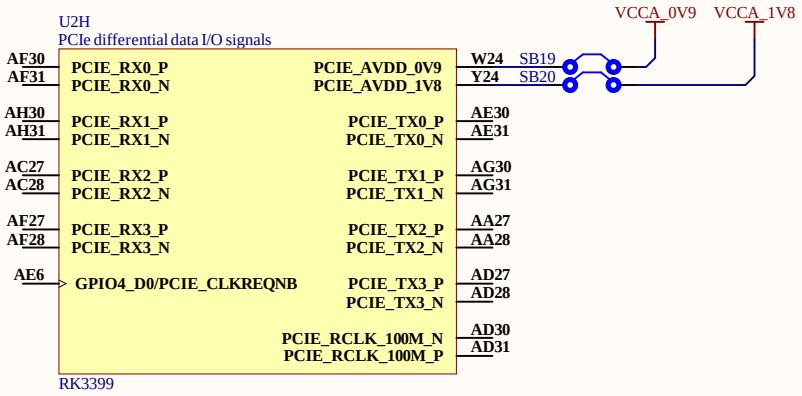




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Size	Number		Revision
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File:	C:\Faizan\..\MISCIO_Sheet.SchDoc	Drawn By:	Huzaifa Ali

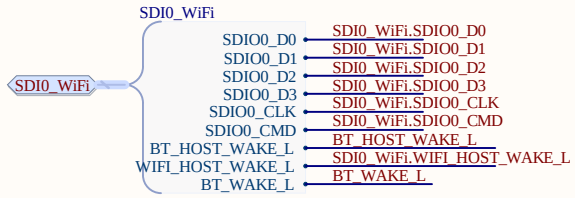
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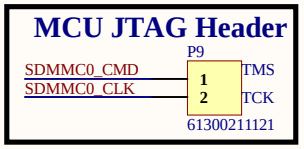


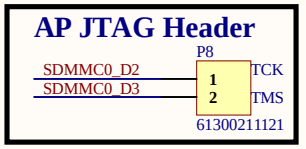
VCCA_1V8 — VCCA_1V8
VCCA_0V9 — VCCA_0V9

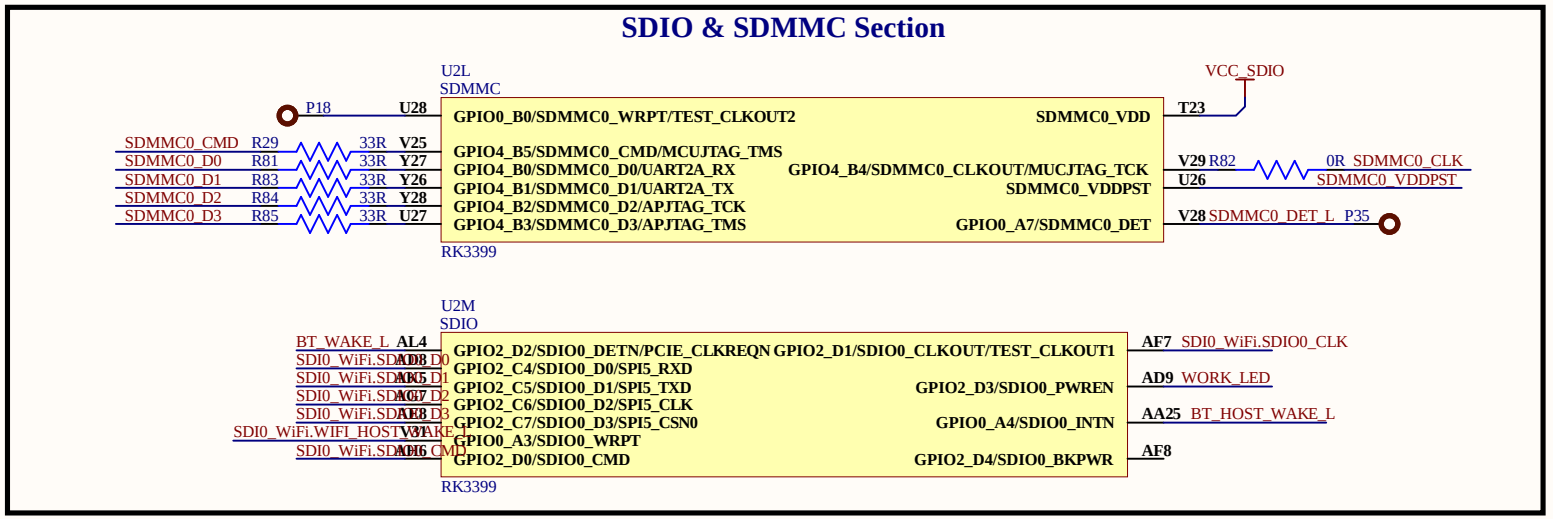
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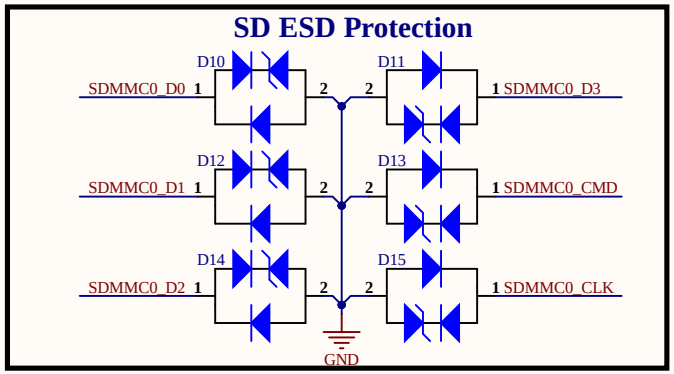
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Size	Number		Revision
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Date:	8/17/2022	Sheet 16 of	21
File:	C:\Faizan\..\PCIe_Sheet.SchDoc	Drawn By:	Huzaifa Ali

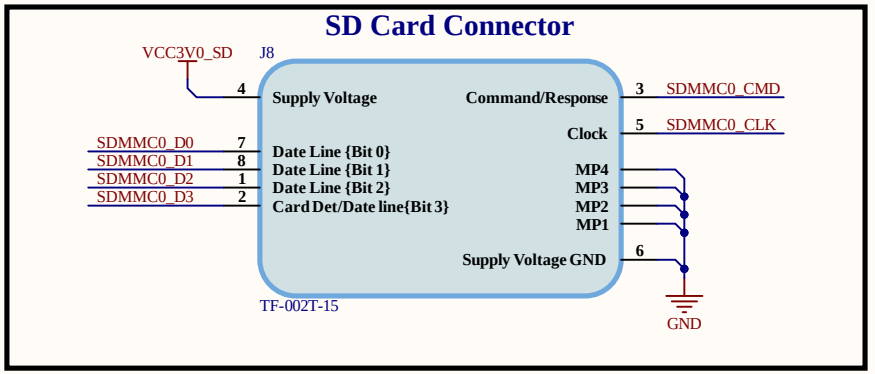


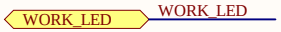


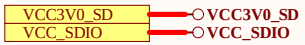






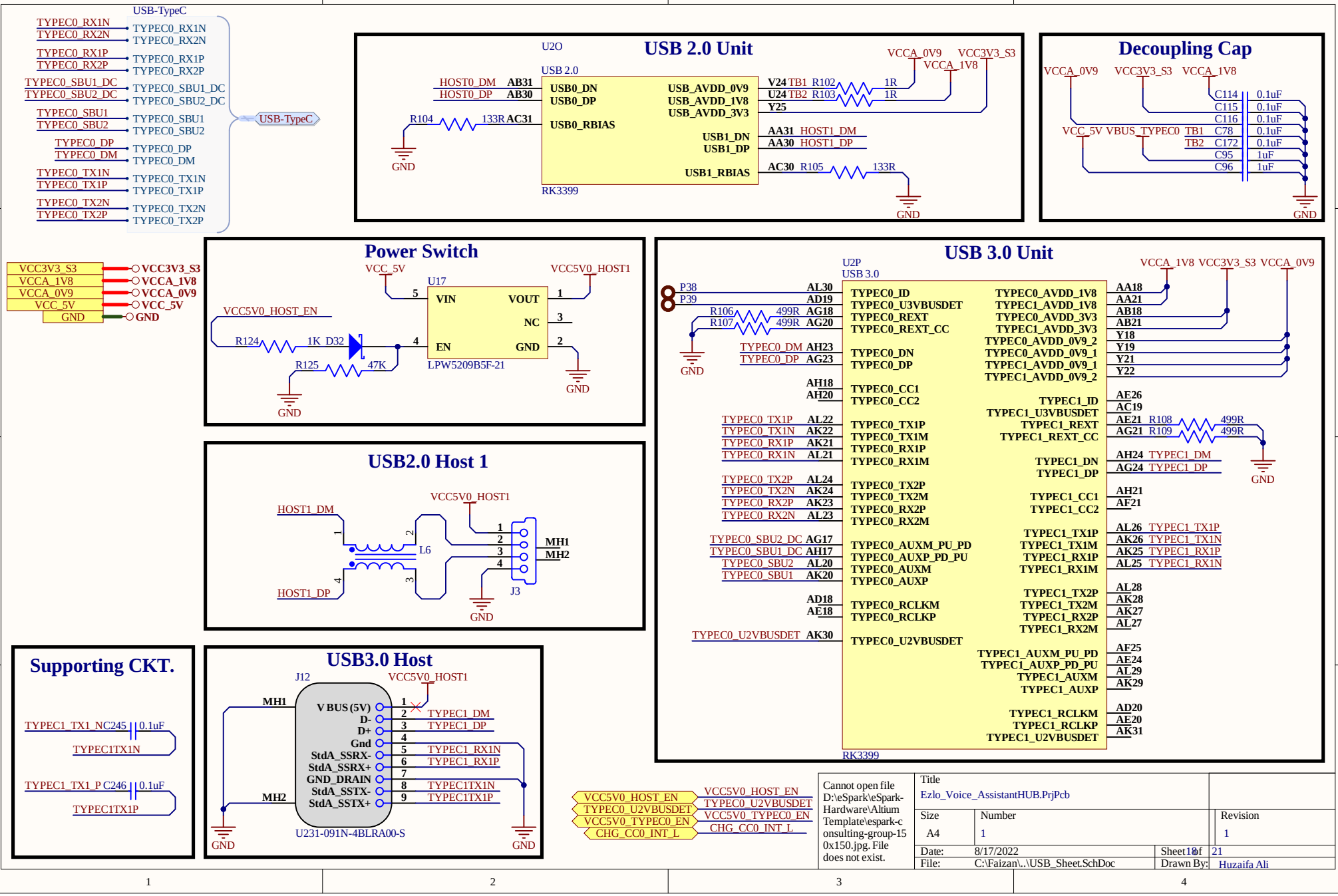


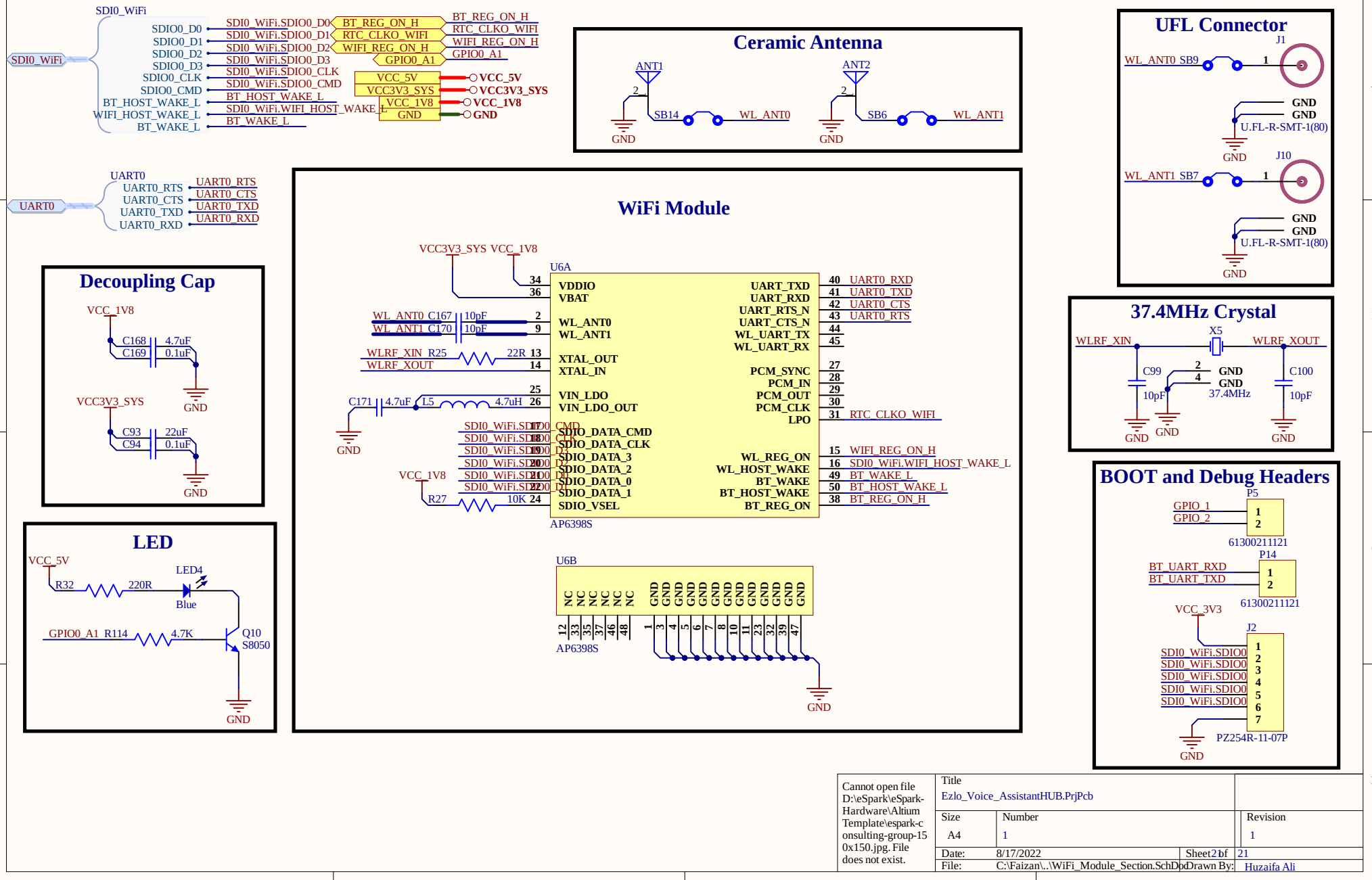


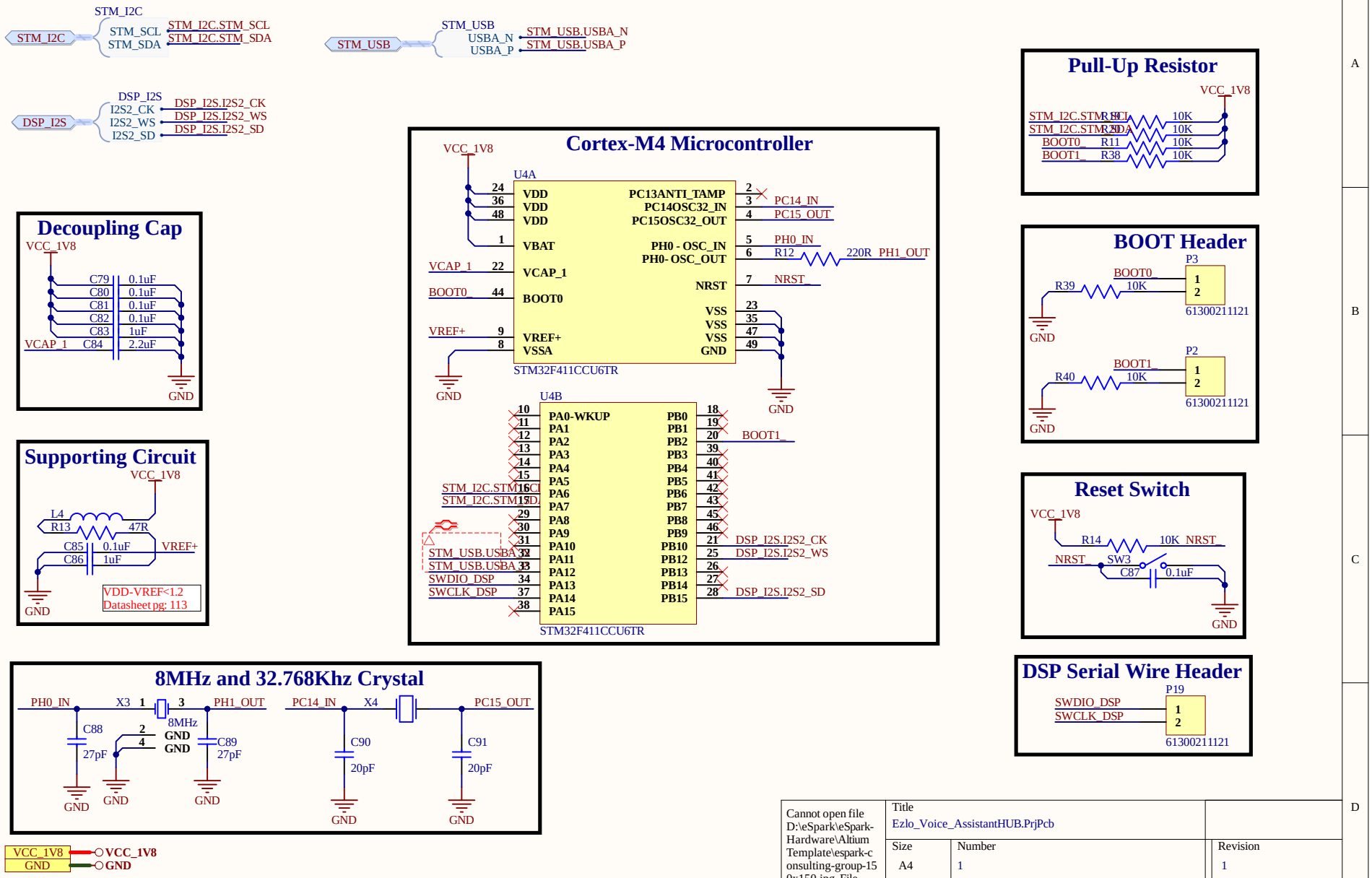


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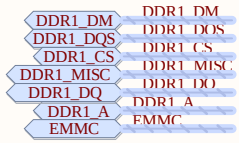
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File:	C:\Faizan\..\SDIO_SDMMC_Sheet.SchDoc	Drawn By:	Huzaifa Ali

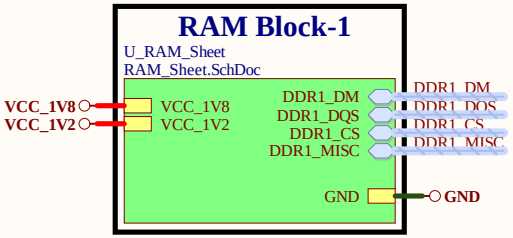

USB-TypeC
TYPEC0_RX1N
TYPEC0_RX2N
TYPEC0_RX1P
TYPEC0_RX2P
TYPEC0_SBU1_DC
TYPEC0_SBU2_DC
TYPEC0_SBU1
TYPEC0_SBU2
TYPEC0_DP
TYPEC0_DM
TYPEC0_TX1N
TYPEC0_TX1P
TYPEC0_TX2N
TYPEC0_TX2P
USB-TypeC
USB 2.0 Unit
U2O
USB 2.0
HOST0_DM AB31
HOST0_DP AB30
USB0_DN
USB0_DP
USB0_RBIAS
R104 133R AC31
USB_AVDD_0V9
USB_AVDD_1V8
USB_AVDD_3V3
USB1_DN
USB1_DP
USB1_RBIAS
V24 TB1 R102 1R
U24 TB2 R103 1R
Y25
AA31 HOST1_DM
AA30 HOST1_DP
AC30 R105 133R
VCCA_0V9
VCC3V3_S3
VCCA_1V8
GND
RK3399
Decoupling Cap
VCCA_0V9
VCC3V3_S3
VCCA_1V8
C114 0.1uF
C115 0.1uF
C116 0.1uF
VCC_5V VBUS_TYPEC0 TB1 C78 0.1uF
TB2 C172 0.1uF
C95 1uF
C96 1uF
GND
VCC3V3_S3
VCCA_1V8
VCCA_0V9
VCC_5V
GND
Power Switch
VCC_5V
VCC5V0_HOST1
U17
VIN
VOUT
NC
EN
GND
LPW5209B5F-21
VCC5V0_HOST_EN
R124 1K D32
R125 47K
USB2.0 Host 1
VCC5V0_HOST1
HOST1_DM
HOST1_DP
L6
J3
MH1
MH2
GND
USB 3.0 Unit
U2P
USB 3.0
P38 AL30
P39 AD19
R106 499R AG18
R107 499R AG20
TYPEC0_DM AH23
TYPEC0_DP AG23
AH18
AH20
TYPEC0_TX1P AL22
TYPEC0_TX1N AK22
TYPEC0_RX1P AK21
TYPEC0_RX1N AL21
TYPEC0_TX2P AL24
TYPEC0_TX2N AK24
TYPEC0_RX2P AK23
TYPEC0_RX2N AL23
TYPEC0_SBU2_DC AG17
TYPEC0_SBU1_DC AH17
TYPEC0_SBU2 AL20
TYPEC0_SBU1 AK20
AD18
AE18
TYPEC0_U2VBUSDET AK30
TYPEC0_ID
TYPEC0_U3VBUSDET
TYPEC0_REXT
TYPEC0_REXT_CC
TYPEC0_DN
TYPEC0_DP
TYPEC0_CC1
TYPEC0_CC2
TYPEC0_TX1P
TYPEC0_TX1M
TYPEC0_RX1P
TYPEC0_RX1M
TYPEC0_TX2P
TYPEC0_TX2M
TYPEC0_RX2P
TYPEC0_RX2M
TYPEC0_AUXM_PU_PD
TYPEC0_AUXP_PD_PU
TYPEC0_AUXM
TYPEC0_AUXP
TYPEC0_RCLKM
TYPEC0_RCLKP
TYPEC0_U2VBUSDET
TYPEC0_AVDD_1V8
TYPEC1_AVDD_1V8
TYPEC0_AVDD_3V3
TYPEC1_AVDD_3V3
TYPEC0_AVDD_0V9_2
TYPEC0_AVDD_0V9_1
TYPEC1_AVDD_0V9_1
TYPEC1_AVDD_0V9_2
AA18
AA21
AB18
AB21
Y18
Y19
Y21
Y22
TYPEC1_ID
TYPEC1_U3VBUSDET
TYPEC1_REXT
TYPEC1_REXT_CC
TYPEC1_DN
TYPEC1_DP
TYPEC1_CC1
TYPEC1_CC2
TYPEC1_TX1P
TYPEC1_TX1M
TYPEC1_RX1P
TYPEC1_RX1M
TYPEC1_TX2P
TYPEC1_TX2M
TYPEC1_RX2P
TYPEC1_RX2M
TYPEC1_AUXM_PU_PD
TYPEC1_AUXP_PD_PU
TYPEC1_AUXM
TYPEC1_AUXP
TYPEC1_RCLKM
TYPEC1_RCLKP
TYPEC1_U2VBUSDET
AE26
AC19
AE21 R108 499R
AG21 R109 499R
AH24 TYPEC1_DM
AG24 TYPEC1_DP
AH21
AF21
AL26 TYPEC1_TX1P
AK26 TYPEC1_TX1N
AK25 TYPEC1_RX1P
AL25 TYPEC1_RX1N
AL28
AK28
AK27
AL27
AF25
AE24
AL29
AK29
AD20
AE20
AK31
VCCA_1V8
VCC3V3_S3
VCCA_0V9
RK3399
Supporting CKT.
TYPEC1_TX1_NC245 0.1uF
TYPEC1TX1N
TYPEC1_TX1_P C246 0.1uF
TYPEC1TX1P
USB3.0 Host
J12
VCC5V0_HOST1
MH1
MH2
V BUS (5V)
D-
D+
Gnd
StdA_SSRX-
StdA_SSRX+
GND_DRAIN
StdA_SSTX-
StdA_SSTX+
TYPEC1_DM
TYPEC1_DP
TYPEC1_RX1N
TYPEC1_RX1P
TYPEC1TX1N
TYPEC1TX1P
U231-091N-4BLRA00-S
GND
VCC5V0_HOST_EN
TYPEC0_U2VBUSDET
VCC5V0_TYPEC0_EN
CHG_CC0_INT_L
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Size
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Number
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Revision
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Date: 8/17/2022
Sheet 18 of 21
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Drawn By: Huzaifa Ali

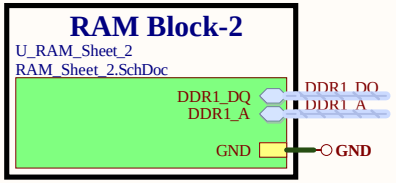

SDI0_WiFi
SDIO0_D0
SDIO0_D1
SDIO0_D2
SDIO0_D3
SDIO0_CLK
SDIO0_CMD
BT_HOST_WAKE_L
WIFI_HOST_WAKE_L
BT_WAKE_L
SDI0_WiFi.SDIO0_D0
SDI0_WiFi.SDIO0_D1
SDI0_WiFi.SDIO0_D2
SDI0_WiFi.SDIO0_D3
SDI0_WiFi.SDIO0_CLK
SDI0_WiFi.SDIO0_CMD
BT_HOST_WAKE_L
SDI0_WiFi.WIFI_HOST_WAKE_L
BT_WAKE_L
BT_REG_ON_H
RTC_CLKO_WIFI
WIFI_REG_ON_H
GPIO0_A1
VCC_5V
VCC3V3_SYS
VCC_1V8
GND
BT_REG_ON_H
RTC_CLKO_WIFI
WIFI_REG_ON_H
GPIO0_A1
VCC_5V
VCC3V3_SYS
VCC_1V8
GND
Ceramic Antenna
ANT1
ANT2
SB14
SB6
WL_ANT0
WL_ANT1
GND
UFL Connector
J1
J10
WL_ANT0_SB9
WL_ANT1_SB7
U.FL-R-SMT-1(80)
UART0
UART0_RTS
UART0_CTS
UART0_TXD
UART0_RXD
WiFi Module
U6A
VCC3V3_SYS
VCC_1V8
VDDIO
VBAT
WL_ANT0
WL_ANT1
XTAL_OUT
XTAL_IN
VIN_LDO
VIN_LDO_OUT
SDIO_DATA_CMD
SDIO_DATA_CLK
SDIO_DATA_3
SDIO_DATA_2
SDIO_DATA_0
SDIO_DATA_1
SDIO_VSEL
UART_TXD
UART_RXD
UART_RTS_N
UART_CTS_N
WL_UART_TX
WL_UART_RX
PCM_SYNC
PCM_IN
PCM_OUT
PCM_CLK
LPO
WL_REG_ON
WL_HOST_WAKE
BT_WAKE
BT_HOST_WAKE
BT_REG_ON
AP6398S
U6B
Decoupling Cap
LED
37.4MHz Crystal
X5
WLRF_XIN
WLRF_XOUT
BOOT and Debug Headers
P5
P14
J2
61300211121
PZ254R-11-07P
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Size
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Number
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Revision
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Date: 8/17/2022
Sheet 2 of 21
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Drawn By: Huzaifa Ali

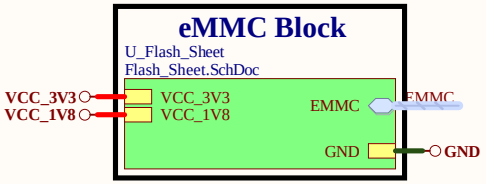


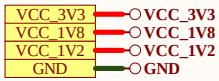
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	Date: 8/17/2022	Sheet 22 of 21	
	File: C:\Faizan\..\DSP_Section.SchDoc	Drawn By: Huzaifa Ali	



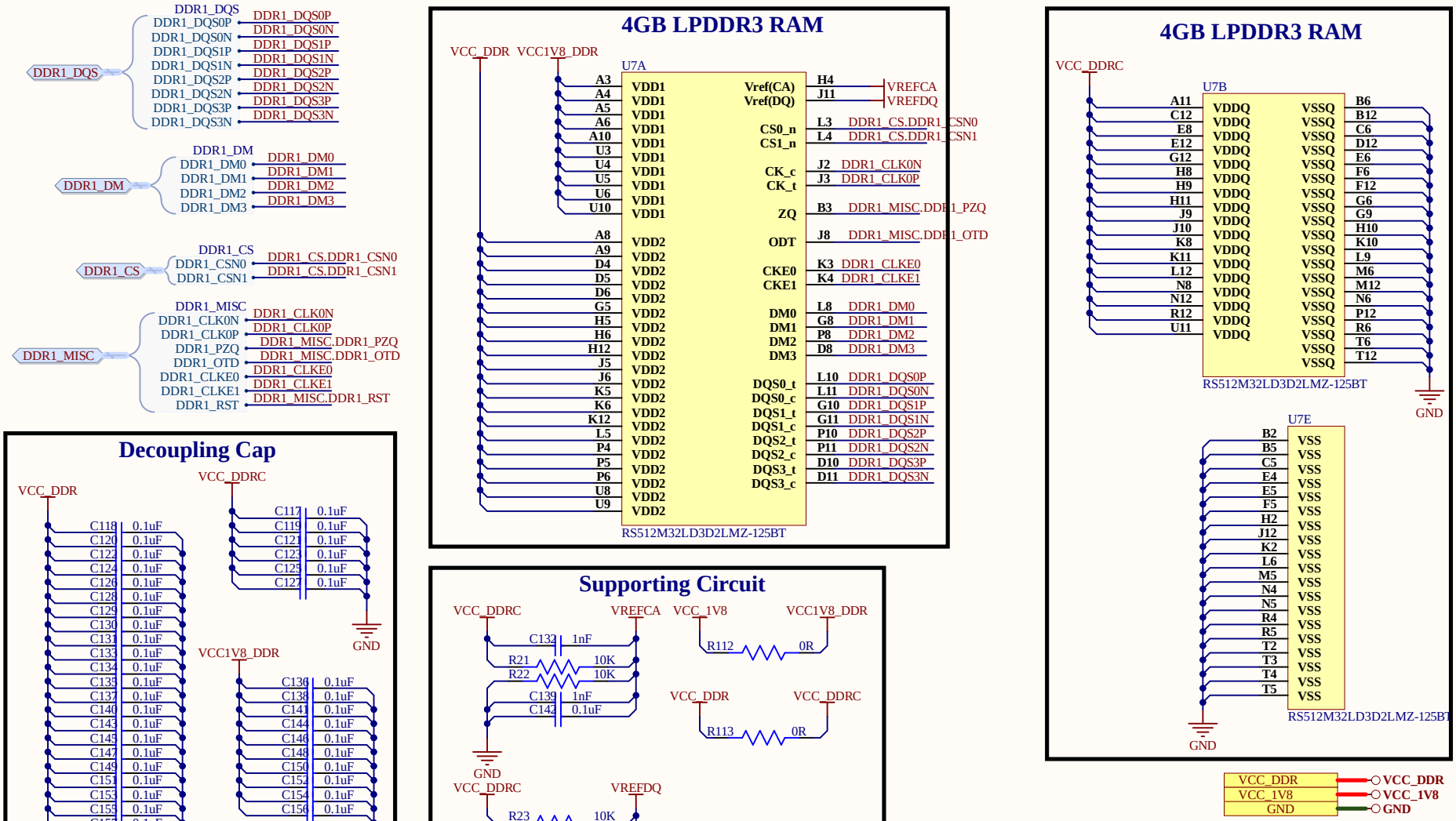




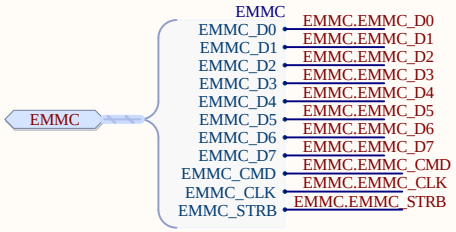


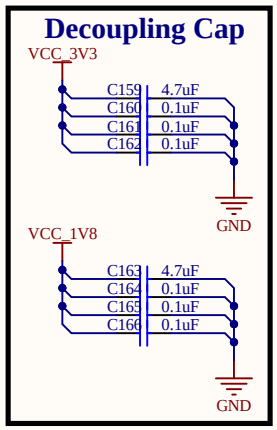


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_AssistantHUB.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/17/2022	Sheet 22 of 21	
	File: C:\Faizan\..\Memory_Section.SchDoc	Drawn By: Huzaifa Ali	

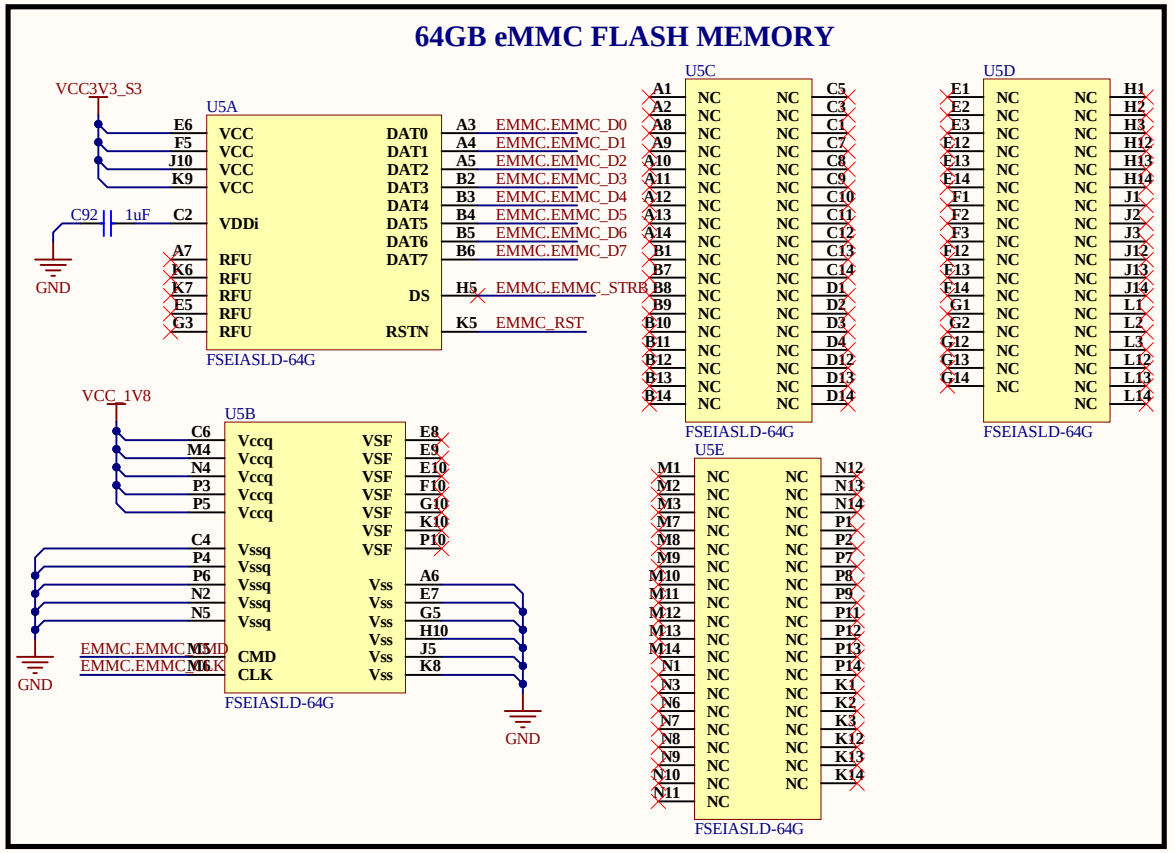


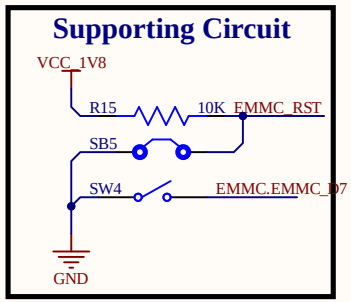
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	Date: 8/17/2022	Sheet 24 of 21	
	File: C:\Faizan\..\RAM_Sheet.SchDoc	Drawn By: Huzaifa Ali	



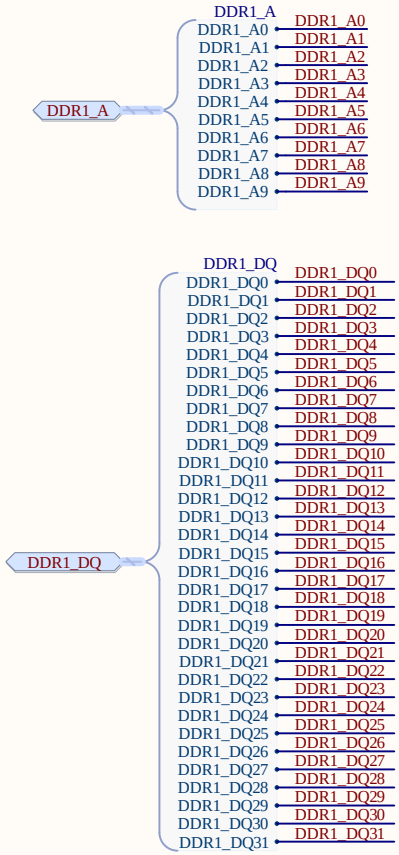


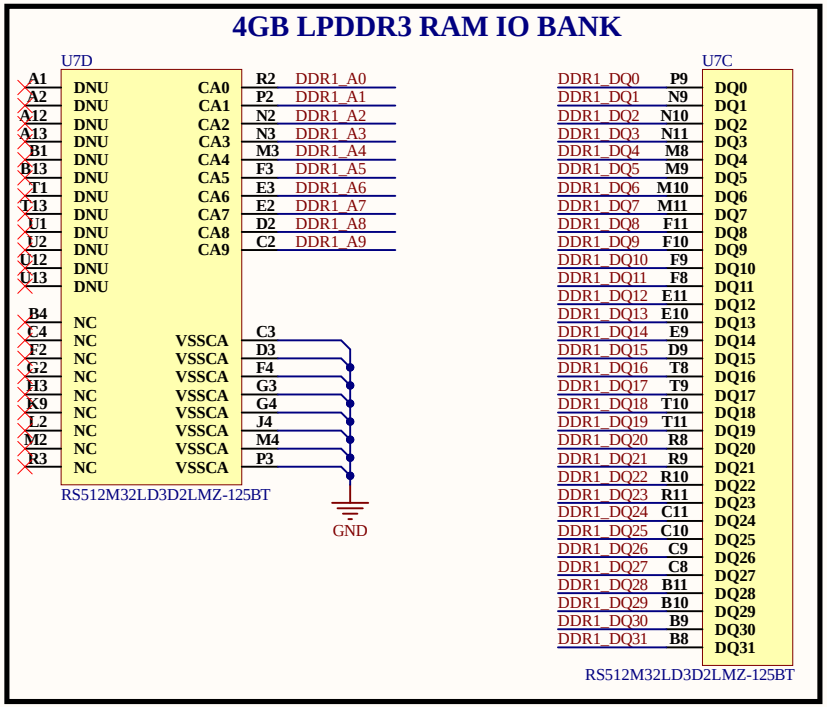






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	Date: 8/17/2022	Sheet 25 of 21	
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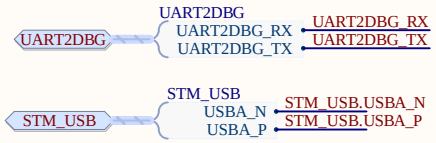




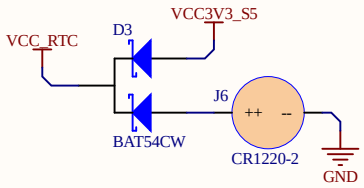


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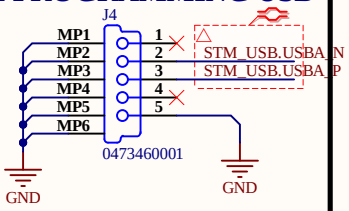
Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	8/17/2022	Sheet 26 of	21
File:	C:\Faizan\..\RAM_Sheet_2.SchDoc	Drawn By:	Huzaifa Ali



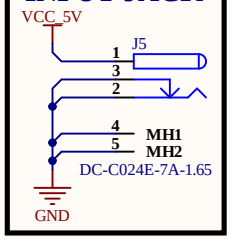
RTC Battery



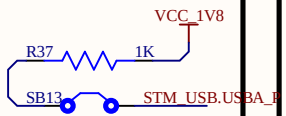
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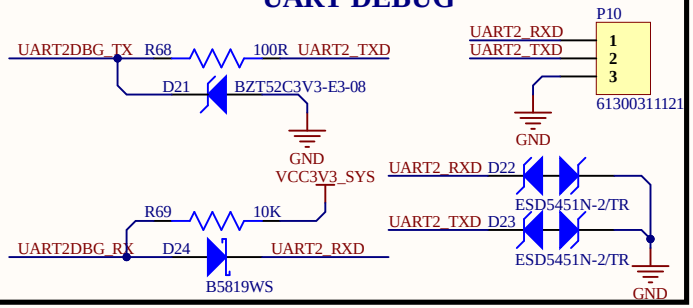
INPUT JACK



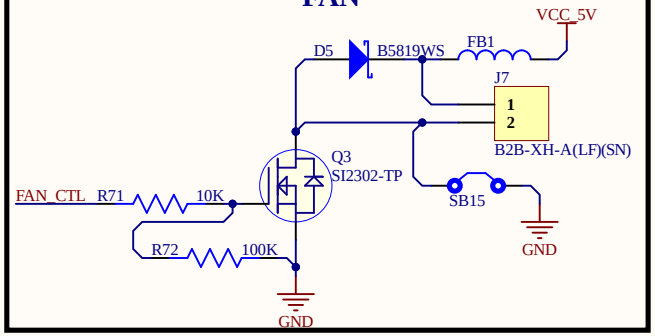
RPULL for Full Speed USB



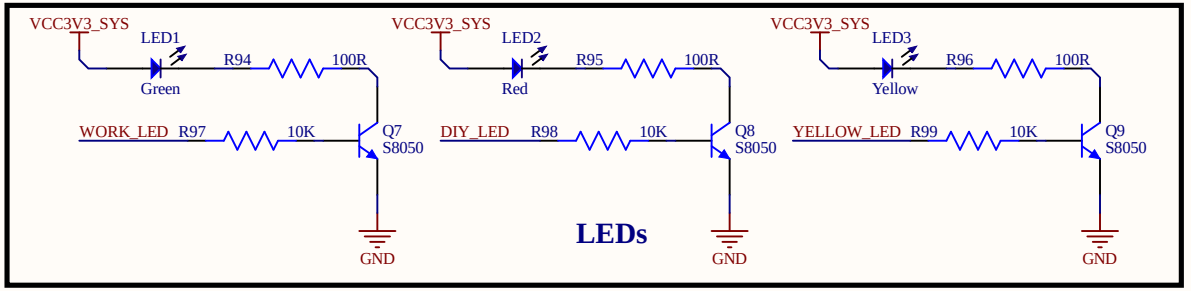
UART DEBUG



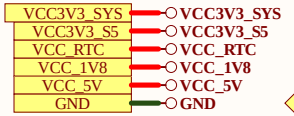
FAN

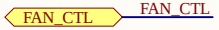


LEDs









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Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	8/17/2022	Sheet 27 of	21
File:	C:\Faizan\..\Connector_Section.SchDoc	Drawn By:	Huzaifa Ali